

BARNAUL, Russian Federation

WMO: 298380

Lat: **53.364N**Lon: **83.539E**Elev: **255**StdP: **98.30**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-30.9	-37.1	0.1	-33.7	-34.0	0.2	-30.6	12.9	-6.7	10.9	-7.4	2.1	330	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.1	18.8	28.4	18.2	26.9	17.6	20.9	26.8	19.8	25.6	18.9	24.9	3.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.2	23.6	17.9	13.2	22.8	16.8	12.4	21.9	61.7	26.8	58.1	25.6	55.0	25.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.9	7.8	-36.9	33.7	4.1	2.1	-39.9	35.2	-42.3	36.4	-44.6	37.5	-47.6	39.0		
(5)			-37.0	23.3	4.1	1.9	-40.0	24.7	-42.4	25.8	-44.7	26.9	-47.6	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-16.8	-13.7	-5.6	4.9	12.6	18.2	19.8	17.6	11.2	4.1	-6.0	-13.1
	DBStd	14.14	8.18	7.34	6.25	5.80	5.39	3.97	3.15	3.68	4.56	5.35	7.80	7.83
	HDD10.0	3616	830	662	483	172	36	2	0	1	39	195	479	716
	HDD18.3	5811	1088	896	742	404	189	50	22	58	217	442	729	974
	CDD10.0	1008	0	0	0	18	116	247	304	237	76	11	0	0
	CDD18.3	163	0	0	0	1	11	46	67	35	4	0	0	0
	CDH23.3	1567	0	0	0	10	164	422	563	349	58	1	0	0
	CDH26.7	396	0	0	0	2	39	100	148	93	14	0	0	0
Wind	WSAvg	3.5	3.4	3.4	3.8	4.0	3.9	3.2	2.7	2.9	3.3	3.8	4.1	3.9
Precipitation	PrecAvg	452	24	18	19	28	39	55	74	47	35	36	41	33
	PrecMax	559	43	39	43	96	99	120	146	97	89	101	89	74
	PrecMin	307	4	1	3	3	9	12	23	13	6	11	13	12
	PrecStd	70	11	10	10	18	20	26	33	20	21	19	18	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	2.0	10.0	24.6	30.2	31.4	32.2	32.8	28.6	21.8	11.0	2.3
		MCWB	0.4	0.1	5.0	12.6	15.9	18.9	19.9	19.9	15.7	11.6	6.1	0.3
	2%	DB	-0.8	0.0	6.3	20.4	27.2	29.2	29.9	29.0	24.8	17.9	8.1	0.7
		MCWB	-1.8	-1.4	2.5	11.0	14.8	17.9	19.9	18.8	14.9	9.9	4.7	-0.4
	5%	DB	-3.1	-1.8	4.2	17.4	25.0	27.7	28.1	27.0	21.8	14.9	5.2	-1.2
		MCWB	-4.0	-3.1	1.5	9.3	13.8	17.7	19.2	17.7	13.5	8.5	2.8	-2.3
	10%	DB	-5.5	-3.5	2.6	14.2	22.2	25.9	26.6	25.0	19.2	12.1	3.0	-3.0
		MCWB	-6.4	-4.7	0.6	7.8	12.9	17.2	18.6	17.0	12.2	7.3	1.4	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.7	5.2	13.6	18.3	21.2	22.7	21.4	17.6	12.3	7.3	0.8
		MCDB	1.3	1.6	9.3	21.8	27.0	27.7	29.3	28.0	24.8	18.6	9.3	1.6
	2%	WB	-1.8	-1.3	3.2	11.5	16.2	19.7	21.3	19.9	15.4	10.5	4.7	-0.5
		MCDB	-0.8	-0.2	5.6	18.9	24.5	26.1	27.0	26.5	22.4	16.5	7.5	0.4
	5%	WB	-4.1	-3.0	1.8	10.0	14.9	18.8	20.4	18.7	14.2	9.1	3.0	-2.3
		MCDB	-3.2	-1.8	3.8	16.6	22.4	25.2	25.7	25.4	20.6	14.2	5.0	-1.5
	10%	WB	-6.4	-4.7	0.6	8.3	13.7	18.0	19.5	17.7	13.0	7.6	1.3	-3.8
		MCDB	-5.5	-3.5	2.3	13.9	20.7	24.3	24.7	23.6	18.2	11.5	2.8	-3.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.7	9.8	10.6	12.1	11.1	10.7	11.4	11.2	9.2	7.1	8.1
		MCDBR	8.2	9.2	10.2	15.5	15.7	13.8	13.2	14.5	15.6	14.6	8.1	8.4
		MCWBR	7.7	8.2	7.5	8.6	7.6	6.2	5.6	6.2	7.6	8.2	5.9	7.9
	5% WB	MCDBR	8.2	9.0	8.8	14.2	13.4	11.8	11.2	12.9	13.6	13.2	7.3	8.3
		MCWBR	7.8	8.2	6.7	8.6	7.4	5.9	5.5	6.0	7.1	7.9	5.6	8.1
Clear-Sky Solar Irradiance	taub		0.295	0.330	0.381	0.408	0.407	0.404	0.429	0.405	0.369	0.355	0.337	0.303
	taud		2.022	1.959	1.905	2.110	2.206	2.295	2.224	2.272	2.320	2.237	2.041	1.960
	Ebn at Noon		630	728	772	816	840	847	815	813	798	710	563	519
	Edh at Noon		77	113	149	139	133	123	130	117	99	86	76	68
All-Sky Solar Radiation	RadAvg		1.10	2.15	3.54	4.63	5.60	6.24	5.64	4.82	3.41	1.78	0.90	0.74
	RadStd		0.09	0.15	0.33	0.26	0.54	0.44	0.32	0.33	0.34	0.25	0.14	0.11